

# Building SGR A\*

Engineering a scientific simulator in an unfamiliar domain: from literature and numerical models  
to an interactive, evidence-backed product

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## Summary

SGR A\* is a browser-based simulation and sonification environment centred on the supermassive black hole at the centre of the Milky Way. It exposes several physical models of increasing fidelity, runs interactively, supports non-visual access, and carries an evidence apparatus intended to let its results be challenged rather than merely admired.

This paper is not a description of the simulator. It is an engineering account of what had to be discovered before the software could be trusted, and of the disciplines that made those discoveries possible.

The central finding is that the hardest part of implementing scientific software was not translating equations into code. It was translating scientific assumptions into explicit software contracts, finding independent ways to test those contracts, and recognising when representation, numerical method or experimental design had invalidated an otherwise plausible comparison.

Every substantive claim below is tied to a dated finding in the project record. Where a result remains provisional or open, it is marked as such. Appendix A tabulates the evidence; Appendix B states what this work does *not* claim.

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## 1. The engineering problem

I chose a problem where plausible output would not be enough.

A black-hole simulator can look convincing while being scientifically wrong. A trajectory can be smooth and visually impressive while using the wrong coordinate representation, the wrong numerical method, the wrong timestep policy or the wrong reference data. Nothing about the rendered image distinguishes those cases.

The requirements were also in tension with one another:

- scientific fidelity sufficient that differences between models mean something;
- numerical stability in strong-field regimes;
- interactive performance in a browser;
- comprehensible interaction;
- non-visual access and sonification;
- evidence strong enough to survive hostile review.

The original product exposed four explicit physics modes: Newtonian; Newtonian with first post-Newtonian (1PN) corrections; the addition of Lense–Thirring spin effects; and adaptive Kerr.

That ladder was initially treated as a progression in fidelity, with each rung expected to own a region of the parameter space where it was the right cost/accuracy trade. One of the later lessons was that an attractive architecture is only a hypothesis until evidence supports it. The work showed that the intermediate rungs did not establish automatic ownership bands: 1PN never did, and automatic Lense–Thirring ownership was retired. The production architecture moved to an automatic Newtonian-to-Kerr treatment, with 1PN and Lense–Thirring retained as explicit comparison and sandbox modes, which is what they were good for.

That revision is representative of the project. The architecture was not defended because it had been designed that way. It was changed when the evidence said the assumptions behind it were weaker than expected.

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## 2. Engineering outside an established domain

My professional background is in commercial systems engineering and, for twenty-four years, in computer science teaching and research supervision. It is not in relativistic dynamics, orbital mechanics, numerical relativity, sonification or auditory display. I could not rely safely on intuition in any of those areas.

That academic background mattered here, but not as a credential. It mattered as a method: read critically, identify assumptions, compare sources, distinguish a claim from the evidence for it, and keep pushing when a result does not make sense.

I could not responsibly ask a model for an equation, receive plausible code, and assume the result was correct. Whenever an implementation or an explanation depended on scientific assumptions I did not understand well enough, I went back to the literature. That happened when orbital behaviour did not match physical expectation; when post-Newtonian and Kerr results disagreed; when different integrations gave contradictory answers; when it became necessary to know which coordinate system a published result actually assumed; and when algebraically valid geodesic constants turned out not all to correspond to physically admissible states.

It happened outside physics too. The sonification and accessibility work began with a broad literature sweep to map the field, followed by a deeper adversarial pass to challenge the first interpretation and extract design constraints. The resulting authority document reviewed twenty-six works with explicit confidence-in-transfer ratings before any audio code was written.

The pattern was consistent across domains:

explore broadly, identify authoritative material, challenge the first interpretation, extract design constraints, implement, then test the resulting behaviour against something independent.

The lesson is not that AI assistance replaces expertise. It is that AI assistance can compress the path from authoritative sources to an executable engineering understanding, provided the engineer retains responsibility for the interpretation and for what counts as closure.

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### 3. The literature is not a software specification

Scientific papers are written to communicate scientific results. They are not written as production specifications. Equations arrive with assumptions. Notation differs between sources. Coordinate systems are often implicit. A closed-form solution may depend on a definition that resembles the one in the product without being the same. Algebraic solutions may admit several roots when only one has physical meaning.

Three cases from this project show three distinct failure modes.

**The published expression was wrong.** A sign discrepancy appeared between Will & Maitra Eq. A3 and the corresponding expression in Hergt & Schäfer. The resolution was not to decide which paper looked more authoritative. It was an independent algebraic cross-check, which established the correct sign from the mathematics rather than from the citation.

**The mathematics underdetermined the answer.** In the geodesic constants work, the solver could produce more than one algebraically valid combination of the relativistic constants  $E$ ,  $L_z$  and  $Q$  for a given set of orbital parameters. Across a 1,400-cell grid, 22 cells admitted two mathematically valid solutions, of which only one corresponded to the intended physical orbit. The literature supplied the mathematics; the software needed something the papers left to expert interpretation, namely an explicit *physical admissibility contract*.

**The published definition was not our definition.** The closed-form solutions for  $(E, L_z, Q)$  available in the literature use an inclination parameter that differs from the inclination definition declared for this project's dataset. They therefore could not be used directly, and the constants had to be solved from the project's own definition. A closed form that solves a slightly different problem is worse than no closed form, because it looks usable.

The general point for any scientific or regulated domain: mathematical validity does not by itself determine product behaviour. The software must encode assumptions the source may never have written down.

## 4. Representation is part of correctness

The most consequential finding in the project came from a result that simply did not make physical sense.

Six orbital states at high eccentricity ( $e = 0.9$ ,  $p = 52.1536$ , profile families G02–G12) were being classified as unbound under the post-Newtonian arms while the Kerr reference classified them as bound. It would have been easy to treat this as one model being a rougher approximation than the other, or to adjust tolerances until the curves agreed. Instead the question became:

Are these calculations actually representing the same physical state in the same way?

They were not.

The implemented 1PN acceleration assumes harmonic (Cook–Scheel) gauge coordinates. The seed states were Kerr–Schild / Boyer–Lindquist states handed to that force law with no gauge transformation applied.

Two features of the diagnosis matter as much as the result.

First, the gauge assumption was *proved from the source*, not inferred from a plausible story. A numerical Christoffel/gauge-family test showed the residuals scaling as  $O((M/r)^2)$  at  $\lambda = 1$  and  $O(M/r)$  at  $\lambda = 0$ , which identifies the gauge family the implemented law belongs to.

Second, the instrument used to investigate was itself validated before conclusions were drawn from it. The production JavaScript force law was transcribed into Python and checked against the original to a worst-case relative error of  $6.28 \times 10^{-16}$  before the probe harness was trusted.

The consequence was not cosmetic. The representation mismatch induced a specific-energy offset of approximately  $3.7 \times 10^{-3} c^2$  against a true binding energy of approximately  $1.82 \times 10^{-3} c^2$  — twice the quantity being measured, and therefore easily sufficient to flip a bound orbit to an unbound classification. After applying the validated Kerr–Schild to harmonic transformation in the correct black-hole spin frame, including the  $20^\circ$  spin-axis tilt, all six disputed states became ordinarily bound, reaching apoapsis within 0.69–0.74 Kerr radial periods.

Nor was the effect confined to the hand-picked cases. A 204-state probe across the affected profile families showed 154 states crossing at least one radial-period contour threshold once the correction was applied, with a median improvement of roughly 57 times. The affected comparison science had to be regenerated.

The engineering lesson is more transferable than the transformation:

**A scientifically correct equation can produce the wrong software result if the representation contract around it is wrong.**

This is the same class of defect that appears when currencies or valuation dates are silently mixed in a financial system, when coordinate reference systems differ in a geospatial pipeline, when timestamp semantics differ across a distributed system, or when two schemas attach different meanings to identically named fields. In each case both sides can be individually correct and the comparison between them still meaningless.

The structural response mattered as much as the fix. Wherever possible, comparison moved away from raw coordinate values and onto quantities that are invariant under the relevant representation change: frequency ratios  $\Omega_\phi/\Omega_r$  and  $\Omega_\theta/\Omega_r$ , and apsidal and nodal advance per radial period. Raw Cartesian comparison between harmonic-gauge post-Newtonian output and Kerr–Schild reference output was prohibited as an error metric.

If two representations differ, do not compare raw values merely because they share units. Compare on quantities whose meaning survives the transformation.

## 5. Discriminating physical, representational and numerical disagreement

Several of the most productive investigations began with the same observation: the result did not look like the physics we expected. The discipline that emerged was to establish *what kind* of disagreement it was before changing anything.

There are at least four candidate explanations for any discrepancy: it is physical (the models genuinely differ); it is representational (§4); it is numerical (the integration is not resolving what it claims to); or it is an ordinary implementation defect. A language model, or a confident engineer, can produce a persuasive explanation for any of the four. The experiment has to discriminate between them.

**A case where the answer was physical.** A gate compared production output against a leading-order analytic model and showed a 1.91% disagreement at 20 AU. The question was whether that was integration error. An independent DOP853 integration of the *same equations of motion* differed from the production trajectory by approximately  $10^{-8}\%$ , seven orders of magnitude below the disagreement under investigation. The 1.91% therefore could not be integration error; it was 1PN cross-talk against a leading-order model. The gate was reclassified as a comparison against an analytic model rather than a numerical-integration gate.

**A case where the answer was numerical resolution, but not convergence failure.** In a strong-field close-passage scenario, systematic timestep refinement produced clean convergence at observed order 2.00–2.01. Separately, the production timestep policy was found to be floor-limited in that same scenario: 89 of 314 periapsis-active steps hit `DT_MIN` exactly, at a worst floor ratio of 0.297, meaning the policy wanted a step roughly 3.4 times smaller than the floor permitted. The careful conclusion was that the scheme converges and the live resolution is capped. A resolution ceiling is not a convergence failure, and conflating the two would have sent the work after the wrong defect.

**A case where the measurement itself was wrong.** This is the sharpest instance in the record. A gate claimed that Lense–Thirring incremental energy drift converged under timestep refinement, and was marked pass on 2% and 0.5% thresholds. Its own output printed `convergence_ratio_dt1_to_dt025 = 1.0000933` — that is, no convergence at all. The number was printed and not acted upon, and the gate’s stated purpose was therefore not met.

The root cause was that `totalEnergy()` computed Newtonian kinetic plus Newtonian potential energy and carried no 1PN conserved-energy correction. With 1PN active, Newtonian energy is not the conserved quantity, so the reported drift was dominated by a physical, timestep-independent offset that had no reason to converge. The corrected experiment nulled the 1PN term, under which Newtonian energy genuinely *is* conserved by Newtonian plus Lense–Thirring dynamics because  $\mathbf{a}_{LT} \cdot \mathbf{v} \equiv 0$ . The residual drift over 30 orbits at  $a = 40$  AU then came out at  $7.8 \times 10^{-12}\%$  at zero spin and  $5.1 \times 10^{-12}\%$  at spin 0.9: round-off, and far stronger evidence than the original claim. The load-bearing guard became the orthogonality identity  $|\mathbf{a}_{LT} \cdot \mathbf{v}| / (|\mathbf{a}_{LT}| |\mathbf{v}|) = 1.4 \times 10^{-16}$ , which is exact.

Three lessons compressed into one finding: a gate can pass while its own diagnostic contradicts it; you cannot measure conservation using a quantity that is not conserved; and the corrected experiment was more useful than the claim it replaced.

**Why the horizon matters.** Throughout, generated analyses offered reassuringly small local error figures. Those numbers answer the wrong question. A small error per substep is not automatically harmless: if it carries a systematic component, thousands of individually negligible errors accumulate into phase drift, secular energy bias, incorrect precession or shifted event timing. The relevant standard is not whether the solver reports a small local error. It is whether the complete numerical method preserves the observable we care about, over the horizon on which we care about it.

## 6. Numerical method is part of the scientific model

It is tempting to treat the integrator as an implementation detail sitting beneath the equations. This project demonstrated otherwise. Step size, minimum step policy, error control, event handling, root refinement and the choice between fixed-step and adaptive methods all determine the regimes in which the simulation can make a reliable claim.

**A very small detail can matter.** One phantom seed event was traced to a negative floating-point denormal in a seeded radial derivative  $dr/dt$ . The value was numerically negligible, but it crossed an event guard that interpreted its sign physically. Nothing was wrong with the intended physics, and nothing was wrong with floating-point arithmetic. The software contract simply failed to distinguish numerical noise from a physically meaningful sign change.

**A structural defect can hide behind a respectable scheme.** The production `physicsStep` is kick-drift-kick with a three-iteration fixed-point refinement for the velocity-dependent 1PN and Lense-Thirring terms. KDK is second-order, symplectic and time-reversible *at constant step size*. With a state-dependent step chosen before each step, both properties are lost: the map is no longer time-reversible, no modified Hamiltonian exists, and energy error becomes secular — growing roughly linearly in step count — rather than bounded and oscillatory. Clamping at `DT_MIN` makes this worse precisely where the dynamics are stiffest. This is the most likely dominant contributor to observed long-run phase drift, and it is an architectural finding rather than a tuning issue.

**A refinement sequence is not automatically a convergence study.** One reported “method envelope” was built from a sequence that does not converge. Refining the dynamic safety factor from 0.002 to 0.001 gave a deviation of  $7.613 \times 10^{-3}$  AU; refining to 0.0005 gave  $5.895 \times 10^{-3}$  AU — a 23% reduction for a factor-of-two refinement, with the two refined levels differing from each other by  $1.718 \times 10^{-3}$  AU, or 29% of the total. In another frame the deviation *grew* by a factor of 2.5 under refinement. Neither sequence is monotone and no order of accuracy is extractable. The mechanism was the floor again: tightening the safety factor cannot reduce the step below `DT_MIN`, so refinement only raises the floor-hit fraction and resamples a different discrete trajectory. Calling the spread between two points on a non-convergent sequence an envelope gives the number a status it has not earned.

This section records the numerical findings at the level needed for the engineering argument. Their mathematics, experimental design and methodology are developed in a companion paper on numerical evidence in scientific software.

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## 7. Building and challenging a scientific oracle

A scientific simulator cannot validate itself by running the same code twice. Independent references are therefore essential: high-order numerical integration of the same equations, specialist relativistic tooling, literature values, invariants where no closed form exists, and deliberately adversarial fixtures.

The discipline that emerged was: establish known cases; compare against an independent implementation or a published result; freeze qualified fixtures; use invariants where no simple closed-form answer exists; perturb inputs and use negative controls; distinguish agreement with the oracle from proof that the product executes the same path; and challenge the oracle itself.

That last clause turned out to be load-bearing.

**The oracle can be confidently, structurally wrong.** In an early convergence study, one moderate-field scenario appeared to plateau at roughly 1% error regardless of refinement. That is the signature of genuine non-convergence, and had the investigation stopped there it would have been reported as such. The cause was a Newtonian-softening gap in the oracle. In a second scenario, the production and reference trajectories disagreed by 55 AU, and the disagreement *grew* under refinement — by the plan’s own criteria, an outright

failure. The cause was a `pnDamp/rSafe` gap in the oracle. After both oracle defects were corrected, both scenarios converged cleanly at observed order 2.00–2.01.

A defective oracle does not produce mild noise. It produces a confident, structured and entirely wrong scientific conclusion, complete with a plausible physical narrative.

**Two reference implementations of the same method had diverged.** The project carried two copies of a DP5(4) integrator, one on the bench/reference side and one on the product side. In a synthetic endpoint test the bench copy overshot its requested endpoint by roughly three orders of magnitude while the product-side copy handled it correctly. The reference implementation was the defective one. That instance was closed; no structural guard against a recurrence — a single-implementation rule — yet exists.

**Agreement with an oracle is not, by itself, evidence about a product.** This was the most important discovery of the 26 August deep audit. Fourteen tranche gate files carried a substantial body of apparently strong evidence for the Kerr lane. That evidence had been produced using an adaptive DP5(4) path. The shipping product was not executing that path: it used its own uncontrolled fixed-step RK4 implementation. The science may well have been correct about the system the harness exercised. It was not evidence about the product.

**Agreement with an oracle is evidence about a product only when the executable path under validation has been established independently.**

The remedy moved the product adapter to bounded adaptive DP5(4) with explicit tolerances, step bounds, subdivision and failure propagation. Implementation gates and bounded validation passed. Final strong-field tolerance ratification remains provisional and is recorded as such.

**Oracle infrastructure needs the same provenance discipline as product code.** A later harness audit found that the Kerr geodesic oracle harness did not check dependency identity, and that ten checkpoints were being silently skipped. The skips were made explicit; that is not the same as proof of full oracle execution, and it is not claimed to be. There is also a physics constraint on that comparison which is easy to get wrong: the reference tooling integrates a *geodesic*, whereas the product's bodies receive mutual half-kicks every sub-step. The comparison is valid only with mutual coupling disabled, as an explicitly single-body configuration. Comparing a coupled run against a geodesic oracle and calling the difference “drift” would be precisely the category error this paper is about.

## 8. An experiment that cannot fail has produced no evidence

The most uncomfortable class of defect found in this project was not in the physics or the solver. It was in the apparatus that was supposed to be checking them.

**An admissibility criterion that could not become true.** A validation experiment compared several tolerance lanes and used their ordering to decide whether a Kerr result was resolved. It returned a null result, which could straightforwardly have been reported as “unresolved”. Closer examination showed something worse: a frozen step ceiling (`hMax = 1`) meant all three lanes were driven by the same effective limit, so the criterion intended to distinguish them was unsatisfiable by construction in all 72 cells. The correct outcome was therefore not unresolved physics but **not evaluated under this experimental design**, with the governing science target left open. A test that cannot fail has not passed.

**A test suite that was green because it was not looking.** Twelve gate files sat outside the release runner's `*.test.mjs` discovery glob, including one gate that was red. The suite reported success throughout. The remedy was to replace pattern-matched discovery with inventory-driven discovery, after which the suite ran 92 of 92 files and 671 of 671 tests.

**A hash gate that could not mismatch.** The frozen-evidence gate regenerated its artefacts before hashing them, so the comparison was structurally incapable of detecting drift in the evidence it was meant to freeze. This finding remains open at the time of writing.

**Harnesses that silently tolerated missing inputs.** A headless loader continued without error when a module file was absent, meaning parity runs could pass with code missing. Parity infrastructure was added; closure is not claimed.

These four share a shape. In each case the apparatus produced a result, the result was reportable, and the result carried no information about the question being asked. Detecting that requires asking a question one step removed from the usual one: not “did the check pass?” but “could this check have failed, and what exactly would have made it fail?”

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## 9. What the product may claim

Severe close encounters impose a hard limit on what any simulator of this kind can honestly claim, and establishing that limit was itself a significant piece of engineering work.

A close passage involving a massive intruder is a chaotic few-body encounter: any perturbation grows exponentially through the passage. The project’s own data shows this cleanly. A Kerr tolerance change of a factor of four produced an endpoint difference of  $2.68 \times 10^{-13}$  AU. A timestep-policy change produced  $2.58 \times 10^{-2}$  AU. That is roughly eleven orders of magnitude of spread in output for perturbations that are nowhere near eleven orders of magnitude apart in input. These are not two components of a method envelope. They are the same sensitivity sampled at two amplitudes, one small enough to stay on the same discrete trajectory and one not.

The structural consequence is unavoidable:

No two schemes will agree pointwise at the end of a severe encounter, and agreement would not prove correctness if they did.

Scientific credibility in that regime therefore has to rest on invariants and statistics rather than on state distance: energy and angular-momentum error growth, mass-shell residual behaviour, orbital-element distributions, periapsis precession rates. The defensible user-facing claim for severe events is **statistical fidelity, not trajectory fidelity**.

That is a weaker claim than “this is where the star will be”. It is also a claim the simulator can actually support and defend, and stating it explicitly is worth more than an impressive claim that a competent reviewer can dismantle in ten minutes. The same principle governs the accessibility claims in §11 and the open items in Appendix B.

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## 10. Performance without weakening the physics

The easiest way to make a difficult physical model faster is to run less of the difficult physical model. That was ruled out. The working question became: how much of the cost is inherent in the physics, and how much is accidental?

**Some slowness is inherent in the method.** Near the Kerr separatrix, orbital behaviour is genuinely expensive to resolve by local numerical integration: zoom-whirl motion means the radial period diverges logarithmically with distance from the separatrix, as  $T_r \sim -\ln(p - p_{\text{sep}})$ . Within an integration architecture, some increase in work near critical states is therefore legitimate and must not be removed by silently changing the model. That cost belongs to integrating the motion rather than to the physics itself: for an isolated test body the motion admits a closed-form solution whose evaluation cost does not grow near the separatrix, which §12 identifies as a question for V2. There is also an evidential consequence: if whirl count is confounded with solver inefficiency in the timing data, conclusions drawn from wall-clock cost near critical states are confounded until the two are separated.

**Most of it was not.** Instrumentation showed that the timestep-selection routine `pickDt` cost between 2.6 and 5 times as much as the force calculation itself, flat across body counts, and consumed 3,997 ms, or 11.3%, of the measured frame budget. The cause was scope rather than the inner loop: the pairwise requirement was recomputed from scratch every substep across the entire population when only the close body's requirement had actually changed. Moving the comparison into squared space gave a 3.7 times improvement on `pickDt` and 1.9 times end-to-end, with identical physics. The architectural conclusion is that a block scheduler should recompute only the woken block.

Related findings: in the Kerr lane, ten bodies were taking exactly one forced DP5(4) step per substep, an inherited cadence with no defensible justification, with readmission cost of 1.63 million calls scaling directly with it. Endpoint bisection in event refinement was consuming a large number of nested solves; safeguarded Brent refinement reduced that sharply while preserving the event contract. And the accretion-disc renderer costs approximately 11 ms in the Schwarzschild treatment against approximately 473 ms for full Kerr, which is why fidelity tiering had to be a design axis rather than an afterthought.

**The measuring instrument was itself a defect.** The vm-based performance harness inflated measured physics cost by between 8.8 and 18.4 times, and did so non-uniformly. A harness that is uniformly wrong is an inconvenience; one that is non-uniformly wrong reorders the optimisation priorities and sends the work after the wrong bottleneck. It was replaced with a non-vm measurement loader.

The sequence that generalises:

measure with an instrument you have validated; determine whether the cost is physical or accidental; optimise only the accidental part; then re-check the scientific result.

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## 11. Non-visual access: when the literature overturns the design

The initial assumption was that sonification would be the route by which a blind or low-vision user accessed the simulation. The literature review overturned that, and the resulting determination is one of the clearest examples in the project of research changing a design rather than decorating it.

**Sonification is not the accessibility strategy. The structured non-visual rendering of the model is.** The evidence points one way: sonification delivers overview and trend rather than values or geometry (Sharif et al., 2025); untrained listeners obtain little from naive mappings on hard cases (Tucker Brown et al., 2022); the auditory channel carries a false-positive bias (Guiotto Nai Fovino et al., 2024); and no published work establishes audio-only interpretation of trajectory data at all.

The practical build-order rule that followed: if the parallel semantic layer ships and the audio layer does not, the product is accessible. If the audio layer ships and the semantic layer does not, the product is inaccessible with an interesting demo attached. If effort has to be cut, it is cut in that order.

The design decisions that followed are all traceable to published findings rather than to taste:

- A parallel DOM, following the established PhET Interactive Simulations pattern, with landmark regions, a strict heading hierarchy, native HTML controls and description that updates in response to interaction. The canvas is one rendering of a model that has non-visual structure, not the model itself.
- A single audio authority. All sound — sonification, event cues, self-voicing speech, legend playback, confirmation tones — passes through one arbitration layer owning priority, gain and rate. No component emits sound directly.
- Mono as a hard requirement, with no information carried spatially. Assistive technology may split or force mono channels, and listeners arriving from this field read spatial auditory dimensions as kinematics, so a pan intended as azimuth is heard as velocity.
- Loudness carries no data. Neuhoff, Kramer & Wayand demonstrated bidirectional pitch/loudness interference with measurable distortion of the represented data. Loudness is reserved for user volume, focus emphasis and ducking.

- Derivatives live in timing, not pitch contour. Fan et al. (2026) found that the continuous glissando which feels most “scientific” was the worst-performing condition for exactly the judgements it was intended to support.
- Polarity is declared and spoken. There is no natural polarity for “distance to the black hole” (Walker, 2002), and blind and sighted listeners can differ, so the convention is stated in words at the start of every scenario.
- Nothing essential is carried by audio alone, which is both a WCAG requirement and the direct implication of the evidence above.

The legal frame is the Equality Act 2010 anticipatory duty and the updated EHRC Services Code of Practice in force from 5 August 2026, with WCAG 2.2 Level AA as the technical target. The project does not claim certified accessibility, nor usability by people with particular personal circumstances, in the absence of direct user evaluation. That boundary is recorded deliberately.

One further consequence surfaced late and is worth stating generally. Once the interface can change physics modes, select and follow objects, launch intruders and expose runtime state, interface state is no longer merely presentation: it participates in correctness. A control that displays one state while the runtime executes another is an engineering defect, not a cosmetic inconsistency.

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## 12. What the prototype taught V2

V1 was approached as a prototype, and that turned out to be a strength. A prototype reveals constraints that cannot be specified confidently before the system has been exercised.

**Central-body identity.** Identity had accumulated several implicit authorities: array position, role flags, scenario metadata and helper methods. They agreed with one another until scenarios became rich enough to reorder bodies. At that point a perfectly valid ordering could cause an ordinary star to be treated as the central body while Sgr A\* existed elsewhere in the state. The V2 response is stable identity and a single canonical central-body resolver, rather than patching each manifestation as it appears.

**Duplicated integration semantics.** In GR-enabled mode, the block path used a single end half-kick while `physicsStep` used a three-iteration fixed point. Making the loops identical brought the ten-orbit ratio to 1.00 across modes.

**Cheap tools catch what expensive review misses.** Three runtime crashes surviving an earlier AI-directed review were identified by ESLint `no-undef`, which flagged exactly the three undefined identifiers responsible. Sophisticated review and standard static analysis fail differently, and the cost asymmetry between them is enormous.

The same pattern recurred around numerical authority, product-versus-oracle paths, engine-owned time advancement, scheduler semantics, evidence provenance and event handling.

**Which motion needs integrating at all.** The most consequential lesson for V2 came from the performance measurements. Strong-field and close-approach cases force the numerical propagator onto very small timesteps. That is not evidence of incorrect physics; it is the cost of resolving a trajectory by repeated local integration. But Kerr geodesic motion has far more analytic structure than V1 exploits. For an isolated test body, the conserved quantities  $E$ ,  $L_z$  and  $Q$  separate the equations of motion, and closed-form solutions exist. V2 will therefore investigate, on evidence, analytic, semi-analytic and fitted or surrogate representations of test-body motion, retaining numerical integration for interacting, finite-mass and non-geodesic regimes. §3 bears directly on this: the published closed forms use a different inclination definition, and the mapping to the project’s definition for generic spin was not established during V1. An analytic propagator needs only the constants, which V1 already solves from its own definition, so evaluating an orbit does not depend on that mapping; comparing against literature-parameterised tooling does. It also raises a new obligation: if the product adopts the mathematics V1 uses as its oracle, the oracle is no longer independent, and V2 must establish a new reference arrangement before any analytic path is trusted.

This is why V1 is described here as an experimental scientific prototype, validated within its tested domain, rather than presented as a finished scientific instrument. It is not a failed version awaiting repair: its results stand within that domain, and its measurements generated the questions V2 must answer.

V1 answered: *can this be made to work, and what assumptions break when it does?* V2 is designed around a better question: *what must remain true for the system to stay correct as it grows?*

A standing preservation rule follows from this. The diagnostic and provenance infrastructure built during V1 is retained even where it is hidden from the normal interface: its code, telemetry hooks, evidence format, fixtures and re-enablement path are not to be deleted as dead code or made unrecoverable by release packaging. The apparatus that found these defects is not a development artefact to be cleaned up on the way to release.

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## 13. Method

The implementation and investigation were AI-assisted throughout. The working method is treated in detail in the companion paper, *When Green Tests Aren't Enough*; what matters here is the part that shaped the engineering.

Models were used adversarially rather than authoritatively. One model proposes; another challenges; the disagreement is resolved by literature, mathematics, code or experiment; and the corrected understanding becomes part of the engineering contract. The value did not come from any single model being strongest. It came from different models making *different* mistakes, which is considerably more useful.

Two process rules earned their place. First, a technically interesting answer is not necessarily progress on the task: sophisticated, genuinely novel analysis can still be the wrong work, and long investigations need a separate context whose job is to hold the objective, remember what has been ruled out, and redirect drift. Second, closure is on evidence, not on confidence. A persuasive explanation is not a result.

Neither rule is specific to AI. Both become more necessary when the cost of producing a fluent, plausible, wrong answer falls to nearly zero.

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## 14. Why this matters beyond scientific software

None of these lessons requires the next system to contain a black hole. The physics made failures easier to expose, because the domain is unforgiving and has independent references. The underlying problems are general.

A representation mismatch in orbital mechanics is a semantic mismatch between schemas, currencies, time zones or units.

An unreliable scientific oracle is an unreliable golden dataset, a mis-specified reference environment, or a compliance baseline nobody has re-derived.

Long-horizon numerical bias is an individually negligible business-rule error accumulating across millions of transactions.

A gate that cannot fail is a monitoring alert whose threshold can never be crossed, a test suite whose discovery pattern silently excludes files, or a control that samples only the cases it was designed around.

A chaotic close encounter is any system whose outputs are genuinely sensitive to inputs, where the honest product claim is statistical rather than pointwise, and where promising pointwise behaviour is a liability.

An interface state that participates in correctness is any system where presentation and execution can diverge and nobody has decided which is authoritative.

The habits that transfer are decomposition, explicit contracts, independent validation, evidence discipline and a willingness to state the limits of a claim. What AI assistance changed is the range of domains in which those habits can be applied, not the need for them.

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## 15. Conclusion

SGR A\* became more than the simulator I set out to build. It became a practical investigation into how an experienced engineer can operate in a technically unfamiliar, multidisciplinary environment without outsourcing judgement.

The most valuable discoveries were not that equations could be implemented or code generated. They were that correctness lives at the boundaries:

between one coordinate representation and another;

between an equation and its numerical realisation;

between local accuracy and accumulated bias;

between a product and its oracle;

between literature and executable contract;

between an implementation and the evidence for it;

between a test that passed and a test that could have failed.

The project repeatedly produced plausible results that became interesting only when challenged. The habit that mattered most was therefore not asking for better answers. It was learning to ask:

**What would prove that this answer is actually the right one?**

The simulator is one substantial worked example of that method. The method is what I intend to carry into the next domain.

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## Appendix A: Evidence record

Findings are listed by date. “Gate state” records what the existing evidence apparatus reported at the time.

| Date   | Area                          | Gate state                                  | What was actually true                                                                                                 | Remedy / status                                                                                               |
|--------|-------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 5 Aug  | Oracle softening gap          | Scenario appeared non-convergent at ~1%     | Newtonian-softening gap in the oracle                                                                                  | Fixed; order 2.00 recovered                                                                                   |
| 5 Aug  | Oracle<br>pnDamp/rSafe gap    | 55 AU disagreement growing under refinement | Oracle defect, not product failure                                                                                     | Fixed; order 2.00–2.01 recovered                                                                              |
| 5 Aug  | Timestep floor (S6)           | —                                           | 89/314 periapsis steps at DT_MIN, worst ratio 0.297                                                                    | Resolution ceiling documented; not a convergence failure                                                      |
| 17 Aug | LT energy-drift gate          | Pass on 2% / 0.5% thresholds                | Printed convergence ratio 1.0000933, i.e. none; energy measured with a non-conserved quantity                          | Gate withdrawn and replaced; corrected drift $7.8 \times 10^{-12}\%$ / $5.1 \times 10^{-12}\%$ over 30 orbits |
| 17 Aug | 1PN analytic gate             | 1.91% disagreement at 20 AU                 | Same-equation DOP853 differs by $\sim 10^{-8}\%$ ; disagreement is physical                                            | Reclassified as analytic-model comparison                                                                     |
| 19 Aug | Headless loader               | In routine use                              | Continued silently when a module file was absent                                                                       | Parity infrastructure added; closure not established                                                          |
| 19 Aug | Performance harness           | In routine use                              | vm harness inflated physics cost 8.8–18.4x, non-uniformly                                                              | Non-vm measurement loader                                                                                     |
| 26 Aug | Kerr product path             | 14 tranche gate files green                 | Evidence from adaptive DP5(4); product shipped fixed-step RK4                                                          | Product moved to bounded adaptive DP5(4); strong-field ratification provisional                               |
| 26 Aug | Duplicated DP5(4)             | Tranche evidence frozen                     | Bench/oracle copy overshot endpoint $\sim 1000x$ in a synthetic case; shipped copy correct                             | Instance closed; no single-implementation guard                                                               |
| 26 Aug | Release runner                | Release suite green                         | 12 gate files outside the *.test.mjs glob, including one red gate                                                      | Inventory-driven discovery: 92/92 files, 671/671 tests                                                        |
| 26 Aug | Frozen-evidence gate          | Gate present                                | Regenerated its artefacts before hashing them                                                                          | <b>Open</b>                                                                                                   |
| 26 Aug | Timestep cost                 | —                                           | pickDt 2.6–5x computeAccel, flat across body counts; 3,997 ms / 11.3% of frame                                         | 3.7x on pickDt, 1.9x end-to-end, identical physics                                                            |
| 29 Aug | Gauge / representation        | Six states classified OUT - BOUND_CONFIRMED | 1PN law in harmonic gauge fed KS/BL seeds; offset $3.7 \times 10^{-3} c^2$ vs binding energy $1.82 \times 10^{-3} c^2$ | KS-to-harmonic transform in spin frame; all six bound; 154/204 states crossed a threshold, median 57x         |
| 31 Aug | Kerr tolerance experiment     | Null result                                 | Frozen hMax=1 made lanes non-independent; criterion unsatisfiable in all 72 cells                                      | Reclassified as not evaluated, step-ceiling dominated; science target left open                               |
| Aug    | Geodesic constants            | Algebraically valid solutions               | 22 of 1,400 grid cells admitted two valid roots, one physically admissible                                             | Explicit physical admissibility contract                                                                      |
| Aug    | Published expression          | —                                           | Sign discrepancy between Will & Maitra Eq. A3 and Hergt & Schäfer                                                      | Resolved by independent algebraic cross-check                                                                 |
| 8 Sep  | Chaotic-encounter sensitivity | “Method envelope” framing                   | 4x tolerance change: $2.68 \times 10^{-13}$ AU; timestep-policy change: $2.58 \times 10^{-2}$ AU                       | Claim restricted to statistical fidelity for severe events                                                    |
| 9 Sep  | Kerr geodesic oracle harness  | Oracle-backed gates                         | Dependency identity unchecked; 10 checkpoints silently skipped                                                         | Skips made explicit; full-execution proof not claimed                                                         |
| 11 Sep | Non-visual access             | Sonification assumed to be the access route | 26-work review: audio gives trend, not values or geometry                                                              | Parallel DOM made primary; sonification supplementary                                                         |
| 13 Sep | Body interaction (GR on)      | Existing gates green                        | Block path used one end half-kick; physicsStep used a three-iteration fixed point                                      | Loops made identical; 10-orbit ratio 1.00 across modes                                                        |
| 22 Sep | Runtime crashes               | Survived earlier AI-directed review         | ESLint no -undef flagged exactly the three responsible identifiers                                                     | Standard lint adopted in the loop                                                                             |

## Appendix B: Claims not made, and open items

This work does **not** claim:

- **Pointwise trajectory fidelity through severe close encounters.** See §9. The supported claim is statistical fidelity, evidenced through invariants and distributions.
- **Ratified strong-field tolerances for the Kerr lane.** The product path was moved to bounded adaptive DP5(4) and passed bounded validation; final strong-field ratification remains provisional.
- **Proof of full oracle execution** in the Kerr geodesic harness. Silent checkpoint skips were made explicit; that is a narrower statement.
- **Closure on frozen-evidence integrity.** The frozen-evidence gate regenerated artefacts before hashing them and is open at the time of writing.
- **Closure on headless-loader parity.** Infrastructure was added; closure is not established.
- **A structural guard against duplicated numerical implementations.** The one known divergence was closed; no single-implementation rule is yet enforced.
- **Systematically bounded long-run drift.** The symplecticity loss under state-dependent stepping (§6) is identified but not yet remediated, and the long-horizon drift experiment is outstanding.
- **Certified accessibility**, or usability by people with particular personal circumstances, in the absence of direct user evaluation.
- **That the reviewed relativistic literature was used as implementation authority** beyond the specific expressions and definitions named in the text.

Items carried forward: single-implementation guard for numerical methods; frozen-evidence gate repair; long-horizon drift study under a time-reversible or regularised scheme; strong-field tolerance ratification; block-scheduler recomputation scope; direct user evaluation of the non-visual layer; and a V2 investigation of analytic, semi-analytic and fitted propagation of test-body motion, with an independent reference arrangement for it.